

Optimal Block-Circulant Preconditioner: Closed-Form Solution for $\min_C \|I - C^{-1}T\|_F^2$

1 Problem Setup

Let $T \in \mathbb{C}^{N \times N}$ be a block matrix with $N = pn$, where $p = 2m + 1$, consisting of $p \times p$ blocks each of size $n \times n$. We write

$$T = [T_{jk}]_{j,k=0}^{p-1}, \quad T_{jk} \in \mathbb{C}^{n \times n}.$$

We seek a block-circulant matrix C that solves

$$\min_C \|I - C^{-1}T\|_F^2, \quad (1)$$

where C is constrained to have the block-circulant eigendecomposition

$$C = (F \otimes I_n) \Lambda (F^* \otimes I_n), \quad (2)$$

with F the $p \times p$ unitary DFT matrix and $\Lambda = \text{blockdiag}(\Lambda_0, \Lambda_1, \dots, \Lambda_{p-1})$, each $\Lambda_j \in \mathbb{C}^{n \times n}$.

2 Reduction via Unitary Invariance

Set $U = F \otimes I_n$, which is unitary. Define the block-DFT transform of T :

$$\hat{T} = U^* T U = (F^* \otimes I_n) T (F \otimes I_n). \quad (3)$$

We partition $\hat{T}$ into $n \times n$ blocks:

$$\hat{T} = [\hat{T}_{jk}]_{j,k=0}^{p-1}, \quad \hat{T}_{jk} \in \mathbb{C}^{n \times n}.$$

Since $C^{-1} = U \Lambda^{-1} U^*$ and the Frobenius norm is unitarily invariant,

$$\begin{aligned} \|I - C^{-1}T\|_F^2 &= \|U^*(I - C^{-1}T)U\|_F^2 \\ &= \|I - \Lambda^{-1}\hat{T}\|_F^2. \end{aligned} \quad (4)$$

Since Λ is block-diagonal, the (j, k) -block of $\Lambda^{-1}\hat{T}$ is $\Lambda_j^{-1}\hat{T}_{jk}$. Expanding the squared Frobenius norm block-by-block:

$$\|I - \Lambda^{-1}\hat{T}\|_F^2 = \sum_{j=0}^{p-1} \sum_{k=0}^{p-1} \|\delta_{jk} I_n - \Lambda_j^{-1}\hat{T}_{jk}\|_F^2. \quad (5)$$

3 Decoupling over Eigenvalue Blocks

Since Λ_j appears only in the terms with block-row index j , the global optimization (??) decouples into p independent subproblems. For each $j = 0, \dots, p-1$:

$$\min_{\Lambda_j} \|I_n - \Lambda_j^{-1} \hat{T}_{jj}\|_F^2 + \sum_{k \neq j} \|\Lambda_j^{-1} \hat{T}_{jk}\|_F^2. \quad (6)$$

We reparametrize by $B_j = \Lambda_j^{-1}$, converting each subproblem into an unconstrained least-squares problem in $B_j \in \mathbb{C}^{n \times n}$:

$$\min_{B_j} \|I_n - B_j \hat{T}_{jj}\|_F^2 + \sum_{k \neq j} \|B_j \hat{T}_{jk}\|_F^2. \quad (7)$$

4 Consolidation and Normal Equations

We consolidate all blocks in row j of $\hat{T}$ into a single matrix. Define

$$R_j = [\hat{T}_{j0} \ \hat{T}_{j1} \ \cdots \ \hat{T}_{j,p-1}] \in \mathbb{C}^{n \times pn}, \quad (8)$$

and the corresponding target

$$E_j = [0 \ \cdots \ 0 \ I_n \ 0 \ \cdots \ 0] \in \mathbb{C}^{n \times pn}, \quad (9)$$

where I_n occupies the j -th block-column position. Then (??) is equivalent to

$$\min_{B_j} \|E_j - B_j R_j\|_F^2. \quad (10)$$

This is a standard matrix least-squares problem $\min_X \|D - XA\|_F^2$. Taking the derivative with respect to B_j :

$$\frac{\partial}{\partial B_j} \|E_j - B_j R_j\|_F^2 = -2 (E_j - B_j R_j) R_j^*.$$

Setting this to zero yields the normal equation:

$$B_j R_j R_j^* = E_j R_j^*. \quad (11)$$

Expanding the Gram matrix

The Gram matrix of block-row j is

$$G_j := R_j R_j^* = \sum_{k=0}^{p-1} \hat{T}_{jk} \hat{T}_{jk}^*. \quad (12)$$

Note that G_j is precisely the (j, j) diagonal block of the product $\hat{T} \hat{T}^*$. It is Hermitian positive semi-definite by construction.

Expanding the right-hand side

Since E_j picks out the j -th block-column of R_j^* :

$$E_j R_j^* = \hat{T}_{jj}^*. \quad (13)$$

Thus the normal equation (??) becomes

$$B_j G_j = \hat{T}_{jj}^*. \quad (14)$$

5 Closed-Form Solution

Theorem 1 (Optimal eigenvalue blocks). *Assume that for each $j = 0, \dots, p-1$, the Gram matrix $G_j = \sum_{k=0}^{p-1} \hat{T}_{jk} \hat{T}_{jk}^*$ is invertible. Then the unique minimizer of (??) over block-circulant matrices of the form (??) is given by*

$$B_j^* = \Lambda_j^{*-1} = \hat{T}_{jj}^* G_j^{-1} = \hat{T}_{jj}^* \left(\sum_{k=0}^{p-1} \hat{T}_{jk} \hat{T}_{jk}^* \right)^{-1} \quad (15)$$

for $j = 0, \dots, p-1$. If additionally each $\hat{T}_{jj}$ is invertible, the optimal eigenvalue blocks are

$$\Lambda_j^* = G_j (\hat{T}_{jj}^*)^{-1} = \left(\sum_{k=0}^{p-1} \hat{T}_{jk} \hat{T}_{jk}^* \right) (\hat{T}_{jj}^*)^{-1} \quad (16)$$

and the optimal preconditioner is

$$C^* = (F \otimes I_n) \text{ blockdiag}(\Lambda_0^*, \dots, \Lambda_{p-1}^*) (F^* \otimes I_n).$$

Proof. The decoupling in Section 3 reduces the problem to p independent subproblems of the form (??). The compact form is a matrix least-squares problem whose unique solution (when G_j is invertible) satisfies the normal equation (??), giving $B_j^* = \hat{T}_{jj}^* G_j^{-1}$. Inverting yields $\Lambda_j^* = (B_j^*)^{-1} = G_j (\hat{T}_{jj}^*)^{-1}$. $\square$

6 Special Case: Block-Toeplitz T

When T is block-Toeplitz with generating blocks $\{T_{-m}, \dots, T_m\}$, the block-DFT transform $\hat{T} = U^* T U$ is block-diagonal:

$$\hat{T} = \text{blockdiag}(\hat{T}_{00}, \hat{T}_{11}, \dots, \hat{T}_{p-1,p-1}),$$

where each diagonal block is

$$\hat{T}_{jj} = \sum_{\ell=-m}^m T_\ell \omega^{j\ell}, \quad \omega = e^{-2\pi i/p}.$$

Since $\hat{T}_{jk} = 0$ for $j \neq k$, the Gram matrix simplifies to $G_j = \hat{T}_{jj} \hat{T}_{jj}^*$, and (??) gives

$$B_j^* = \hat{T}_{jj}^* (\hat{T}_{jj} \hat{T}_{jj}^*)^{-1} = \hat{T}_{jj}^{-1},$$

so $\Lambda_j^* = \hat{T}_{jj}$. In this case, C^* coincides with the block-circulant matrix whose eigenvalue blocks are the DFT of the Toeplitz generating sequence—recovering the classical result.

7 Remarks

Remark 1 (Geometric interpretation). The optimal $B_j^* = \hat{T}_{jj}^* G_j^{-1}$ is the solution to the oblique projection (in the Frobenius-norm sense) of the identity onto the range of the j -th block-row of $\hat{T}$, restricted to the diagonal block. When $\hat{T}$ is nearly block-diagonal, $B_j^* \rightarrow \hat{T}_{jj}^{-1}$.

Remark 2 (Singular and ill-conditioned cases). If G_j is singular, the minimum-norm solution is $B_j^* = \hat{T}_{jj}^* G_j^\dagger$. When $\hat{T}_{jj}$ is singular, one uses $\Lambda_j^* = G_j (\hat{T}_{jj}^*)^\dagger$.

Remark 3 (Computational cost). The dominant cost is computing $\hat{T} = U^* T U$, requiring $O(p^2 n^2)$ operations (or $O(p \log p \cdot n^2)$ via FFTs on block-columns). Forming each G_j costs $O(p n^3)$ and the inversion $O(n^3)$, for a total of $O(p^2 n^2 + p n^3)$.

Remark 4 (Relation to known preconditioners). This result generalizes the T. Chan optimal circulant preconditioner (which minimizes $\|C - T\|_F$ for scalar circulants) to the block setting with the objective $\|I - C^{-1} T\|_F$. The two objectives yield the same C^* when T is block-Toeplitz, but diverge for general T .

8 Summary of the Algorithm

1. Compute $\hat{T} = (F^* \otimes I_n) T (F \otimes I_n)$ and extract the $n \times n$ blocks $\hat{T}_{jk}$.
2. For each $j = 0, \dots, p-1$, form the Gram matrix $G_j = \sum_k \hat{T}_{jk} \hat{T}_{jk}^*$.
3. Solve for the optimal eigenvalue blocks:

$$\Lambda_j^* = \left(\sum_{k=0}^{p-1} \hat{T}_{jk} \hat{T}_{jk}^* \right) (\hat{T}_{jj}^*)^{-1}.$$

4. Assemble the preconditioner: $C^* = (F \otimes I_n) \text{blockdiag}(\Lambda_0^*, \dots, \Lambda_{p-1}^*) (F^* \otimes I_n)$.